ANALYSIS ON SUPERMARKET 2011 DAILY SALES TRANSACTIONS

In this report, I have examined and analysed a dataset containing the items bought in a supermarket with each transaction with transaction ID. It also includes the time and date of each transaction. The Dataset originally contain 39,194 transactions with 44 columns, out of which 41 columns is for each set(s) items bought during a particular transaction which each row represents. While the other columns was the “Transaction ID” , “Date “ and “Time”

In the Preprocessing of the dataset, a new dataset of 22094 transaction from the original dataset was selected using python selection code while still having same the 44 columns. In the datasets the “time” columns was split in to hour and minutes and these create an addition column of “Hour” to the datasets. This is to done to be able to visualize the time of the where transactions are high, and from the analysis it is discovered that most of the transaction done between the hours 9 am in the morning and 5pm in the in evening and more transaction is carry out around 12noon and 2pm .

Therefore for us to determine the association rules for the items , I employed the ” APRIOR” algorithm with minimum support=0.01, minimum confidence=0.2, minimum lift=3, minimum length of combination of items =2 to help me determine the Association Rule . To further use this algorithm, the items in the new datasets is put into a basket dataset called “ items_df” this is to enable me visualize the items with highest Support , Confidence and Lift . After putting the items in to the basket, it is discovered that some of the most bought items is the basket are the “WHITE HANGING HEART T-LIGHT HOLDER” and “ASSORTED COLOUR BIRD ORNAMENT.” Items and when the Aprior algorithm was employed to determine the association rule 92 Rules was generated.

However some of the frequently bought items in the supermarket like WHITE HANGING HEART T-LIGHT HOLDER” and “ASSORTED COLOUR BIRD ORNAMENT were not present in the rules generated there items are not present because

1. The items are very popular and they are mostly bought alone and used by many persons and when these persons come to buy these items they don’t

buy other items with them that may be or may not be in the generated rule.

2. The Items are not frequently bought with other items because the items does not usually goes with other items for example SET OF 3 CAKE TINS PANTRY DESIGN', goes with PACK OF 72 RETROSPOT CAKE CASES' for making cakes.

   This means these above items have high support but very low confidence which is lowered than .2 or 20percent which was the confidenc level set for the Aprior algorithm.

One of the most frequently bought items which have high support and high confidence is “SET OF 3 CAKE TINS PANTRY DESIGN “ and “GREEN REGENCY TEACUP AND SAUCER” .

One of the best rule I discover was the “GREEN REGENCY TEACUP AND SAUCER” .

With “ ROSES REGENCY TEACUP AND SAUCER” with a support of 0.25 and confidence of 0.70 and lift of 20.

This means these mix is popular and it is frequently bought in the supermarket Therefore the store keeper of manger of the supermarket should frequently stock these items more and properly do some discounting to the items to attract more sales and profit.

Furthermore, I choose to examine two periods which is winter and summer period.

I discovered that more rules where generated in summer than winter that means customer tends to buy items with other items.

In summer than winter In summer 132 rules was generated while winter 106 rules was generated using the Aprior Algorithm of with minimum support=0.01, minimum confidence=0.2, minimum lift=3, minimum length of combination of items =2.

The items which generated rules in the two period was also similar. The interesting thing is that a particular item line which is the "alarm clock" and it varieties are the items which people buy often and also but with other items in these two periods . Therefore this will help increase sales as the store keeper or manager will know that he needs to get more of the Alarm clock Item and stock them in the store and get more during summer because customer will tend to get them more during summer while shopping for other items.

REFERENCE

Vinita, N. (2023) ‘Workshop-5-dataset.zipData set . Available at https://canvas.wlv.ac.uk/courses/36895/pages/activity-5-dot-1 workshop-5-market-basket-analysis-apriori-rules mining?module_item_id=1429811 (Accessed: 24 February 2023).